

LinearRAG: Linear Graph Retrieval Augmented Generation on Large-scale Corpora

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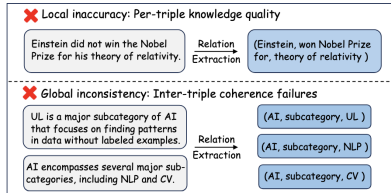
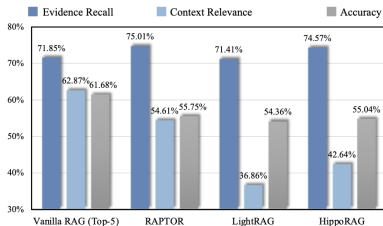
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Outline

- 1 Introduction
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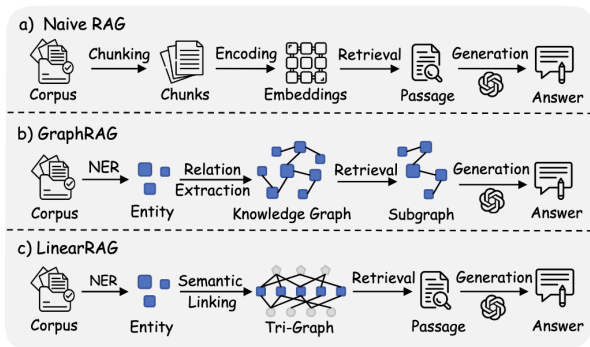
RAG and GraphRAG

- Naive RAG: Effective for simple QA but fails with large-scale unstructured corpora, missing evidence chains.
- GraphRAG: Leverages graph structure to improve recall in retrieving relevant knowledge.



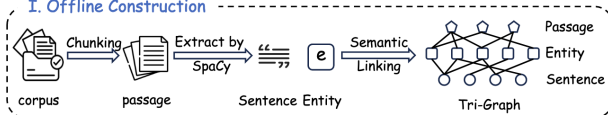
LinearRAG

- Aligned entities serve as anchors connecting passages.
- Contextual relations are preserved within the original passages.

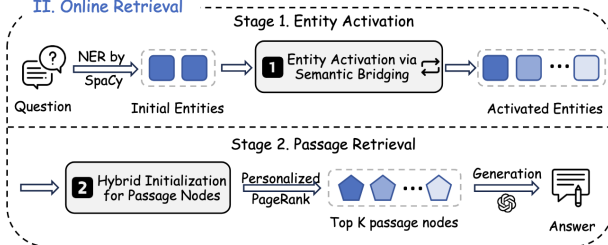


LinearRAG Framework

I. Offline Construction



II. Online Retrieval



Entity activation

1. Initial entity activation

- Entity activation score: (Seed entities E_q)

$$\mathbf{a}_q = [\mathbf{a}_{q,i}]_{|V_e| \times 1}, \quad \text{where} \quad \mathbf{a}_{q,i} = \mathbb{I}_{i=\arg \max_{e_j \in V_e} \text{sim}(e_q, e_j) \cdot \text{sim}(e_q, e_i)}.$$

2. Query-sentence relevance distribution

- Sentence relevance:

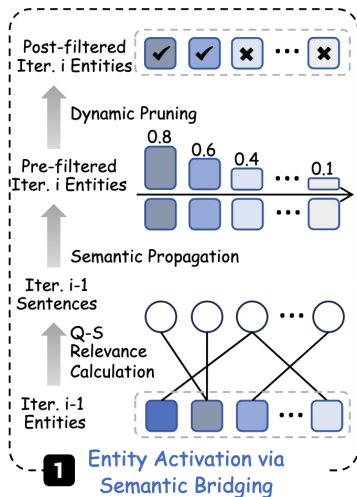
$$\sigma_q = [\sigma_{q,i}]_{|S| \times 1}, \quad \text{where} \quad \sigma_{q,i} = \text{sim}(q, s_i).$$

3. Semantic propagation

$$\mathbf{a}_q^t = \text{MAX}(M^T(\sigma_q \odot (M\mathbf{a}_q^{t-1})), \mathbf{a}_q^{t-1}),$$

4. Dynamic pruning

- Activated entity is retained if its relevance score exceeds threshold δ .



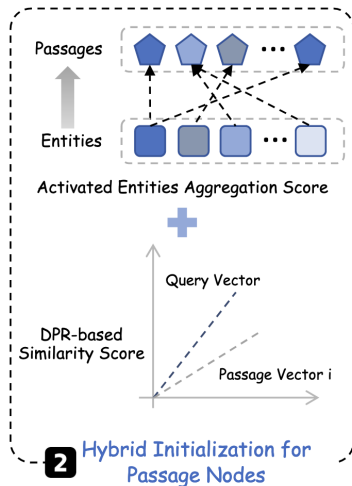
Global importance aggregation

1. Initial score for entity node V_e :

$$I(v_i | v_i \in V_e) = \mathbf{a}_q^{(i)}, \text{ where } \mathbf{a}_q = (\mathbf{a}_q^{(1)}, \mathbf{a}_q^{(2)}, \dots)$$

2. Passage node V_p importance score:

$$I(v | v \in V_p) = \left(\lambda \cdot \text{sim}(q, v) + \ln \left(1 + \sum_{e_i \in E_a} \frac{\mathbf{a}_q^{(i)} \cdot \ln(1 + N_{e_i})}{L_{e_i}} \right) \right) \cdot W_p,$$



Result on benchmark

Method	HotpotQA		2Wiki		MuSiQue		Medical
	Contain-Acc.	GPT-Acc.	Contain-Acc.	GPT-Acc.	Contain-Acc.	GPT-Acc.	GPT-Acc.
<i>Direct Zero-shot LLM Inference</i>							
llama-8B	31.10	27.30	33.60	16.20	7.40	8.10	27.31
llama-13B	24.20	16.80	21.90	10.50	3.30	4.40	28.86
GPT-3.5-turbo	33.40	43.20	28.70	31.00	10.30	21.90	45.60
GPT-4o-mini	38.90	40.20	36.30	31.40	13.60	15.80	42.10
<i>Vanilla Retrieval-Augmented-Generation</i>							
Retrieval (Top-1)	46.30	49.10	36.60	31.70	17.80	21.10	48.01
Retrieval (Top-3)	53.00	56.00	44.90	39.70	25.10	27.50	59.07
Retrieval (Top-5)	55.70	58.60	48.60	43.00	26.10	29.60	61.68
<i>Graph-based Retrieval-Augmented-Generation Methods</i>							
KGP	61.50	60.90	31.60	30.00	25.60	30.10	54.22
G-retriever	42.20	40.60	46.60	27.10	14.40	15.50	50.36
RAPTOR	55.90	58.30	50.10	42.10	23.30	27.40	55.75
E ² GraphRAG	61.00	63.90	54.30	38.10	23.80	26.20	58.00
LightRAG	60.30	59.50	55.20	39.00	27.40	28.60	54.36
HippoRAG	57.00	59.30	66.10	59.90	29.30	24.10	55.04
GFM-RAG	62.70	65.60	66.80	59.60	29.90	34.60	56.07
HippoRAG2	62.90	64.30	62.70	55.00	31.00	35.00	60.77
LinearRAG (Ours)	64.30	66.50	70.20	63.70	33.90	37.00	63.72

Efficiency analysis

Method	Time (s)		Token Consumption ($\times 10^6$)		Accuracy
	Indexing	Retrieval (Avg.)	Prompt	Completion	
G-retriever	2745.94	11.487	6.05	2.26	36.85
RAPTOR	1323.57	<u>0.062</u>	0.81	<u>0.03</u>	46.10
E ² GraphRAG	<u>534.60</u>	0.053	<u>0.78</u>	0.08	46.20
LightRAG	4933.22	10.963	35.52	51.16	47.10
HippoRAG	936.00	1.461	3.05	0.98	63.00
GFM-RAG	1202.77	1.211	3.05	0.98	<u>63.20</u>
HippoRAG2	1147.01	1.694	4.98	1.22	58.85
LinearRAG (Ours)	249.78	0.093	0	0	66.95

Ablation study

